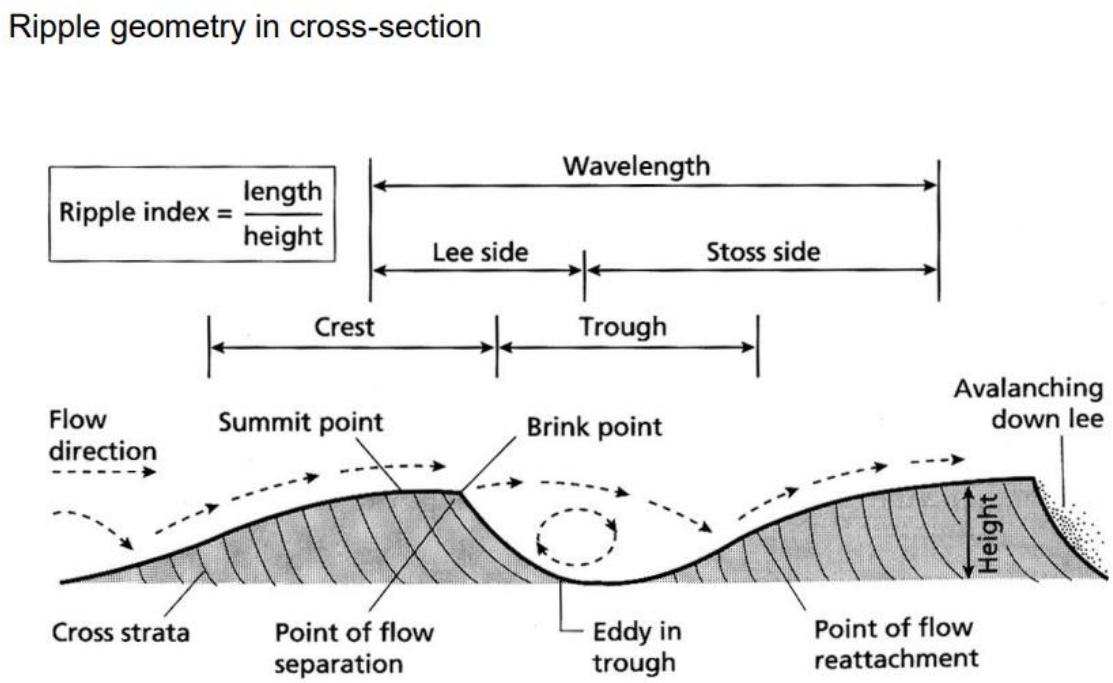
Bedforms and sedimentary structures

Here are some descriptions of the bedforms you will observe in the practical class and details of the sequence of bedforms that are formed as the velocity of the flow increases.

Ripples

Ripples are the very small ridges of sedimentary grains and their longest dimensions, along the crests of the ripples, are at right angles to the current flow



Ripples formed by a unidirectional flow are asymmetrical in cross section with a short, steep lee slope and a longer shallower stoss slope.



Image shows ripple bedforms formed in an experimental flume

https://armfield.co.uk/product/s8-mkii-sediment-transport-demonstration-channel/

The lee slope is always on the down flow part of the ripple. We can use this fact to interpret the direction of ancient flows that deposited ripples.

Ripples can be preserved in the rock record. If only the internal layers are seen these are called cross lamination

Geologists can distinguish ripples formed by the movement of waves from ripples formed by unidirectional currents. The cross sectional shape of ripples formed by the back and forth movements of waves (such as the ripples you might see on a beach) are symmetrical.

Heights: from mm to less than 5-6cm

Wavelengths :less than few tens of cm

Dunes

As the velocity of the flow increase there will be a sudden change to a larger bedform – these are dunes.

Dunes have the same shape as ripples, but they are much larger.

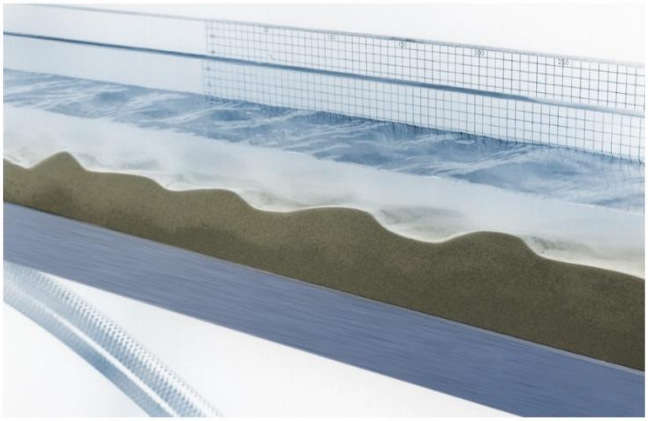


Image shows dune bedforms

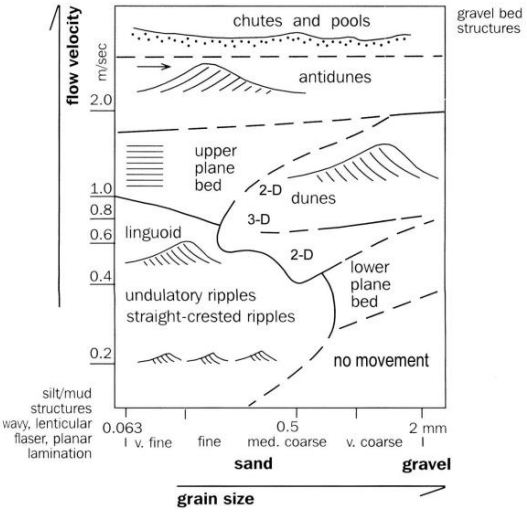
Dunes form in medium sands to gravels under moderate to high flow strengths. They are always asymmetric bedforms that are formed from unidirectional flows only (water or wind).

Heights: several tens of cms up to tens of meters

Wavelengths : Greater than 1m

Cross bedding is the sedimentary structure produced by dunes. It consists of inclined layers within the beds, sometimes inclined up to 35 ^{o}from the horizontal. Commonly it is the steeper lee (down current) slopes of the sand dunes that are preserved.

The sequence of bedforms that are developed under increasing flow conditions



'Bedform phase diagram' showing the stability fields for subaqueous bedforms (and their internal sedimentary structures) resulting from unidirectional flows, in relation to specific velocity-grain size conditions. The flow depth for this diagram is 0.25-0.5 m.